
Project 2 - Utilization of hospital beds during epidemics

Stochastic Simulation

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June 25, 2026

Contents

1	Introduction	1
2	Simulation Model	1
3	Primary Performance Measures	3
4	Sensitivity Analysis	5
5	Discussion	9
6	Use of AI	10

1 Introduction

In this project, we analyze the distribution of hospital beds during a countrywide epidemic, where the hospital has established temporary wards so that it has three distinct wards, Ward A for regular care, Ward B for intensive care, and Ward C for other patient arrivals. Over a simulated 365-day period, patient arrivals are modelled as Poisson processes. The arrival rates for Wards A and B are dynamic and time-dependent second order polynomials, while Ward C maintains a constant arrival rate. Furthermore, we are given a log-normal Length-of-Stay (LOS) distribution to model the duration for each patient.

We will assume that if an intensive care patient arrives and there are no available beds in Ward B, the patient will be admitted to Ward A, provided there is an available bed there. If Ward A is also full, the patient is relocated to another hospital. Similarly, patients arriving for Wards A or C are immediately relocated to another hospital if there are no available beds.

We begin by assuming that the hospital has 75 beds to distribute across the three wards. Our primary goal is to build a continuous-time stochastic simulation model to evaluate baseline system performance, specifically focusing on mean bed utilization, the probability of wards being full upon arrival, and the total number of patients that are being relocated to another hospital. Then we will perform sensitivity analysis, exploring how different bed distributions, changes in the LOS distribution, and varying the total bed count impact the model.

2 Simulation Model

The primary task was to implement a simulation model of the hospital system over the time interval $t \in [0, 365]$. The hospital consists of three wards: Ward A for regular-care patients, Ward B for intensive-care patients, and Ward C for other patients. The total number of beds is fixed, and a chosen bed distribution determines how many beds are assigned to each ward.

Ward and hospital structure

The simulation was implemented using two main classes: a Ward class and a Hospital class. Each Ward object stores the number of beds in that ward, the current number of occupied beds, the arrival rate function, and the parameters for the length-of-stay distribution. The ward also records statistics such as the number of arrivals, the number of patients relocated, and the accumulated occupied bed-time.

The Hospital class contains the three wards and handles the rules for patient flow between them. This is useful because some decisions depend on more than one ward. In particular, intensive-care patients are first sent to Ward B, but if Ward B is full, they may overflow to Ward A. Therefore, this routing rule belongs naturally to the hospital-level simulation rather than to an individual ward.

Patient types and ward assignment

In the implementation, each arrival is associated with a patient type rather than simply with a physical ward. Regular-care patients have Ward A as their default ward, intensive-care patients have Ward B as their default ward, and other patients have Ward C as their default ward. This distinction is important because the physical ward occupied by a patient is not always the same as the patient type. In particular, an intensive-care patient may be admitted to Ward A if Ward B is full.

Therefore, relocation probabilities are recorded by patient type, while bed utilization is recorded by the ward where patients are actually placed. For example, an intensive-care patient who overflows to Ward A still counts as an intensive-care arrival, but contributes to the occupancy and utilization of Ward A.

Arrival processes

The arrivals were simulated as Poisson processes in continuous time. Ward C has a constant arrival rate $\lambda_3 = 6$, so arrivals to Ward C were generated using exponential interarrival times with mean $1/\lambda_3$.

Ward A and Ward B have time-dependent arrival rates. Ward A has arrival rate

$$\lambda_1(t) = -\frac{1}{3650}t^2 + \frac{1}{10}t,$$

and Ward B has arrival rate

$$\lambda_2(t) = \frac{1}{5}\lambda_1(t).$$

Since these rates change over time, arrivals to Ward A and Ward B were generated using thinning. The idea is to first generate candidate arrivals from a homogeneous Poisson process with rate $\lambda_{\max}$, where $\lambda_{\max}$ is an upper bound on the true arrival rate $\lambda(t)$ over the interval $[0, 365]$. Each candidate arrival at time t is then accepted with probability

$$p(t) = \frac{\lambda(t)}{\lambda_{\max}}.$$

If the candidate is accepted, it is included as an actual arrival; otherwise, it is discarded.

For Ward A, $\lambda_1(t)$ is a concave parabola on $[0, 365]$, with maximum at $t = 182.5$. Therefore, we used

$$\lambda_{\max,A} = \lambda_1(182.5) = 9.125.$$

Since Ward B has arrival rate $\lambda_2(t) = \frac{1}{5}\lambda_1(t)$, its maximum rate is

$$\lambda_{\max,B} = \frac{1}{5} \cdot 9.125 = 1.825.$$

These choices ensure that $\lambda(t)/\lambda_{\max}$ is always between 0 and 1, so it can be used as an

acceptance probability.

Length of stay and discharge events

When a patient is admitted, a length of stay is sampled from the relevant lognormal distribution. The discharge time is then computed as

$$t_{\text{discharge}} = t_{\text{arrival}} + \text{LOS}.$$

This discharge time is added to an event queue. The simulation processes both arrivals and discharges in chronological order, so beds are released at the correct time before later arrivals are handled.

The event queue is implemented as a priority queue. This means that the next event processed is always the event with the smallest time. This is important in the continuous-time simulation because arrivals and discharges do not only happen at integer-valued days.

Admission and relocation rules

For each arrival, the simulation checks whether the relevant ward has an available bed. A regular-care patient is admitted to Ward A if possible; otherwise, the patient is relocated. An intensive-care patient is first admitted to Ward B if possible. If Ward B is full, the patient is admitted to Ward A if Ward A has an available bed. Only if both Ward B and Ward A are full is the intensive-care patient relocated. Other patients are admitted to Ward C if possible and otherwise relocated.

This means that the number of relocated patients is counted by patient type. For Ward B, the simulation also records how many intensive-care patients overflowed to Ward A.

3 Primary Performance Measures

With our simulation model, we implemented an initial evaluation using a baseline bed allocation configuration, where $(B_A = 20, B_B = 10, B_C = 45)$.

To capture the performance of the simulated hospital network, we establish four primary performance measures: Probability Full on Arrival, Patient Relocation Probability, Mean Physical Bed Utilization, and Mean Annual Relocated Patients. We explicitly track the distinction between "Full on Arrival" and final "Relocation" probabilities because it reveals how often a specific ward runs out of physical space versus how often a patient is entirely turned away from the hospital. Furthermore, computing the mean annual relocated patients provides the direct baseline required to evaluate system-wide capacity shortfalls and establish a clear target for minimization.

The resulting primary performance metrics expose critical systemic bottlenecks and highlight what we should change in terms of bed allocation to minimize the count of relocations due to the shortage of beds in wards.

The architecture of Wards A and C provides a baseline indicator of pure capacity constraints, as both operate under the same rule-based logic. The simulation results confirm that the blocking probability matches the relocation probability identically for both wards (64.20% for Ward A, 30.69% for Ward C), where if a patient arrived to either Ward A or C, and no beds are available, the patient will be considered as lost.

Even though there are similarities in the way Wards A and C are structured, their failures to accommodate patients given the baseline allocation is caused by different reasons and are showcased in 1.

Table 1: Baseline Simulation Performance Measures (100 Replications, Lognormal LOS)

Performance Metric	Ward A (Regular)	Ward B (Intensive)	Ward C (Other)
Physical Beds Allocated	20	10	45
Mean Probability Full on Arrival	65.60%	45.63%	27.91%
Mean Patient Relocation Probability	65.60%	33.35%	27.91%
Mean Physical Bed Utilization	89.38%	78.69%	93.98%
Mean Annual Relocated Patients	1,455.32	148.76	611.63
Mean Annual Overflow (B $\rightarrow$ A)	—	54.69	—
Total System Relocations / Year	2,215.71 $\pm$ 80.53 patients		

Despite maintaining a high mean bed utilization of 89.38%, Ward A was failing to admit nearly two-thirds of its incoming demand, yielding a 65.60% blocking rate. This indicates that a 20-bed allocation is insufficient to absorb the volatile seasonal spikes generated by its non-homogeneous arrival process. Ward C shows a steady-state saturation bottleneck. It operates at a 93.98% mean utilization—the highest in the simulated hospital wards—under a constant arrival stream ($\lambda_C = 6$).

In comparison to these 2, nearly-failing wards, Ward B is facing a different issue. It exhibits a distinct divergence between its probability of being full on arrival (45.63%) and its final relocation probability (33.35%). When Ward B's 10 physical beds are fully occupied, the arrival logic automatically treats Ward A as a secondary safety valve. Over the simulated year, 54 intensive care patients successfully avoided being turned away from the hospital by spilling over into Ward A.

This structural buffer in Ward B protects intensive care patients. But, it simultaneously forces Ward A to use up its own finite bed-days to absorb Ward B's excess demand, compounding Ward A's already severe 65.60% blocking rate. To ensure the statistical integrity and stability of these primary performance measures, the evaluation was executed across 100 independent macro-replications of a 365-day year.

These baseline metrics reveal a structural resource mismatch across the simulated system. The hospital's capacity is unoptimized, leaving Ward C strained at nearly 94% steady-state utilization while Ward B retains a buffer due to its unique overflow capability. To minimize average patient relocations, beds must be reduced from Ward B and added to expand Wards A and C, which serves as the direct objective for the subsequent grid-search optimization framework.

4 Sensitivity Analysis

Optimal bed distribution

To find the most optimal bed distribution across the three wards, the sum of relocated patients was used as the optimization metric - with the smallest sum reflecting the most optimal distribution. Initially, every possible distribution was tested in a simulation, following which the 20 best distributions were selected for further analysis. In the next step, the 20 best distributions were tested in 30 repetitions of the simulation, after which the best distribution was selected as the most optimal bed distribution. Figure 1 below shows the 10 best bed distributions, with the most optimal distribution being "A=46, B=1, C=28":

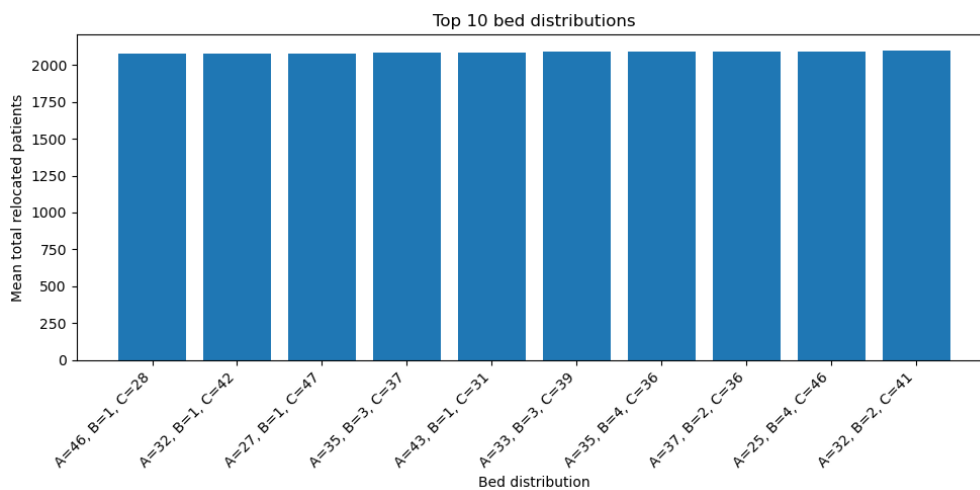


Figure 1: The 10 best bed distributions (from left to right)

As the plot indicates, there were many possible distributions producing very similar relocation sums. This is mainly due to the level of randomness introduced by the generation of arrivals and the LOS, causing slightly different simulation outputs between runs. Therefore, there isn't a singular, most optimal distribution that applies to all simulations. However, there is still an overall trend that can be observed in the plot. Among the best distributions, only very few beds are assigned to Ward B, while the rest are divided more or less equally between Wards A and C. This observation can be explained by the possibility of moving patients from Ward B to A, making Ward B somewhat redundant in this setup. However, in a more realistic hospital setting, having a specialized ward for intensive patients, such as Ward B, is necessary to ensure appropriate patient care. Therefore, the optimization process could be adapted to also minimize the flow of patients from Ward B to A, which would result in an increased bed allocation to Ward B.

After the optimal bed distribution was found, 30 simulations were performed using this distribution, following which the performance measures were determined as described in the previous section. The results from this can be seen in Table 2 below:

Table 2: Simulation Performance Measures With Optimized Bed Distribution (30 Replications, Lognormal LOS)

Performance Metric	Ward A (Regular)	Ward B (Intensive)	Ward C (Other)
Physical Beds Allocated	46	1	28
Mean Probability Full on Arrival	34.60%	93.57%	53.33%
Mean Patient Relocation Probability	34.60%	33.25%	53.33%
Mean Physical Bed Utilization	81.49%	89.10%	96.42%
Mean Annual Relocated Patients	766.10	147.23	1,162.87
Mean Annual Overflow (B → A)	—	267.33	—
Total System Relocations / Year	2,076.20 ± 75.25 patients		

Comparing the results above to the ones from the suboptimal bed distribution described in Table 1, most performance metrics from Wards A and C were improved. Merely the "Mean Annual Relocated Patients" increased for Ward C. However, as the sum of relocated patients was the optimization metric, the "Total System Relocations" still decreased. For Ward B, the measures concerning relocated patients, which includes the relocation probability and annual count, remained almost unchanged. However, all other measures worsened, including the "Mean Annual Overflow", which was more than quadrupled. This is easily explained by the optimized bed distribution only allocating a single bed to Ward B, forcing patients to Ward A, and the optimization criterion only considering the total relocations, disregarding any other performance measures.

Exponentially distributed LOS

To evaluate how the LOS distribution affects the system, the previously log-normally distributed LOS was changed into being exponentially distributed - while still maintaining the same mean.

Initially, 50 simulation runs were performed using the original, log-normally distributed LOS, following which the previously mentioned performance measures were determined. The results of this can be seen in Table 3.

Simultaneously, the same was done using the new, exponentially distributed LOS. The results of this are displayed in Table 4.

Table 3: Simulation Performance Measures With Lognormal LOS (50 Replications)

Performance Metric	Ward A (Regular)	Ward B (Intensive)	Ward C (Other)
Physical Beds Allocated	46	1	28
Mean Probability Full on Arrival	34.42%	93.73%	53.50%
Mean Patient Relocation Probability	34.42%	32.32%	53.50%
Mean Physical Bed Utilization	81.65%	89.65%	96.45%
Mean Annual Relocated Patients	762.20	143.26	1,169.16
Mean Annual Overflow (B → A)	—	272.08	—
Total System Relocations / Year	2,074.62 ± 79.14 patients		

Table 4: Simulation Performance Measures With Exponential LOS (50 Replications)

Performance Metric	Ward A (Regular)	Ward B (Intensive)	Ward C (Other)
Physical Beds Allocated	46	1	28
Mean Probability Full on Arrival	34.29%	93.67%	53.75%
Mean Patient Relocation Probability	34.29%	32.29%	53.75%
Mean Physical Bed Utilization	81.84%	89.42%	96.38%
Mean Annual Relocated Patients	757.74	145.58	1,178.80
Mean Annual Overflow (B → A)	—	276.28	—
Total System Relocations / Year	2,082.12 ± 90.48 patients		

Comparing the performance measures of the two different LOS distributions, it becomes apparent that they are nearly indistinguishable from each other. The difference between the probability percentages can only be observed in the decimal places, while the patient counts differ with single digits. This similarity between the two different distributions can also be observed in Figure 2 below, where the "Mean total relocated patients", which was the optimization metric for the bed distribution, was used as the base of comparison.

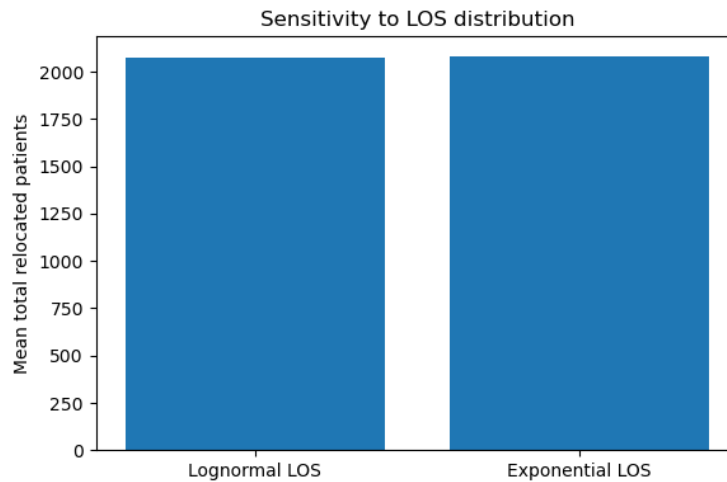


Figure 2: Lognormal LOS vs. Exponential LOS

Impact of total bed capacity

To evaluate the system's sensitivity to the total bed capacity, we expanded our optimization search to simulate total hospital bed allocations of {60, 70, 75, 80, 100}. The goal was to observe how the optimal bed distribution and overall hospital performance metrics scale with changing capacity.

Figure 3 visualizes the mean total number of relocated patients as a function of the total number of beds in the hospital. This illustrates that there is a near-linear inverse relationship between the number of relocations and the number of beds.

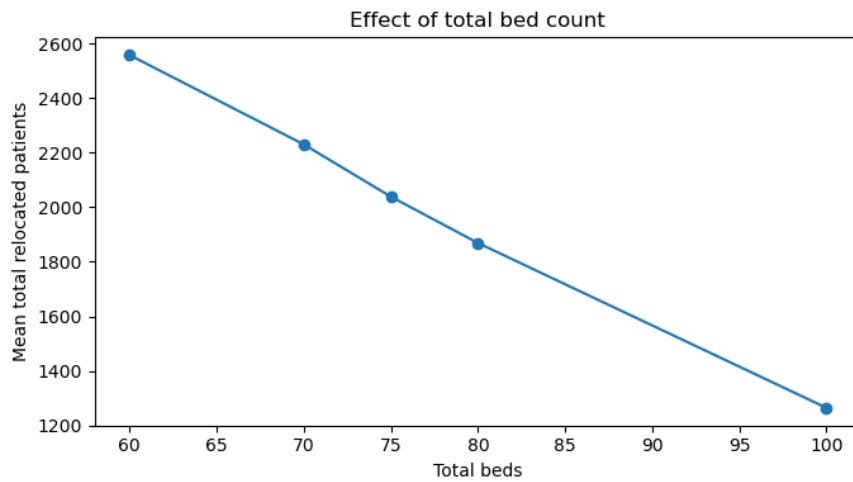


Figure 3: Effect of total bed count on the mean total number of relocated patients.

Furthermore, analyzing how the optimization model allocates these varying capacities reveals a consistent structural strategy. As shown in Figure 4, the allocation to Ward B remains low, at only 1 or 2 beds, regardless of the total capacity, while the remaining beds are distributed somewhat evenly between Wards A and C.

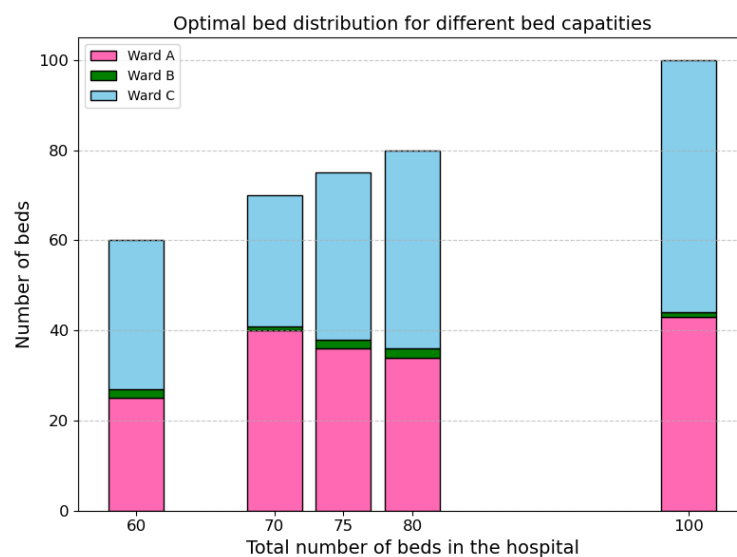


Figure 4: Optimal bed allocation distribution for the varying total bed capacities.

This consistent distribution occurs because of the system's overflow logic. Because an intensive care patient can overflow into Ward A without being counted as a relocation, the analytically optimal solution is to minimize Ward B's bed capacity. This frees up beds to expand Wards A and C. A bed placed in Ward A offers higher systemic flexibility, as it can service both regular care patients and intensive care patients coming from Ward B.

Table 5 shows the mean performance metrics across these configurations.

Table 5: Impact of different total bed capacities on mean hospital performance metrics

Total	Beds			Full Probability			Utilization			Relocated
	A	B	C	A	B	C	A	B	C	
60	25	2	33	59.86%	87.78%	45.59%	88.55%	88.23%	95.91%	2557.15
70	40	1	29	41.52%	93.35%	51.55%	83.90%	89.89%	96.31%	2229.75
75	36	2	37	45.21%	87.58%	39.22%	84.82%	88.36%	95.19%	2038.20
80	34	2	44	47.85%	87.76%	28.67%	85.78%	89.42%	94.03%	1868.50
100	43	1	56	37.59%	93.65%	12.24%	82.90%	88.98%	90.09%	1264.55

This table shows that the probability that all beds in a ward are occupied upon arrival (Full Probability) decreases significantly for Wards A and C as total bed capacity increases, while it remains consistently high for Ward B. For Ward A, the probability that all beds are occupied drops from 59.86% at 60 beds to 37.59% at 100 beds. For Ward C, the drop is even larger, falling from 45.59% at 60 beds to only 12.24% at 100 beds. Similarly, the mean utilization shows that the fraction of occupied beds in Wards A and C decreases as the total number of beds increases, whereas Ward B remains almost constant.

The total number of patients relocated to a different hospital decreases substantially as total capacity expands. Compared to the baseline of 75 beds with 2038.20 relocations, reducing the capacity to 60 beds results in 2557.15 relocations, which is an increase of roughly 25%. Conversely, increasing the total bed capacity to 100 reduces the total number of relocated patients to 1264.55, representing a decrease of almost 38% from the baseline.

Because of the randomness in the optimization algorithm, the resulting optimal bed distribution depends to some extent on the chosen random seed. Therefore, the optimal allocation of beds across the wards for a given total bed capacity can vary between runs.

5 Discussion

The discrepancies encountered across different machines highlights the critical importance of rigorous dependency and environment management in stochastic simulations. While setting a global seed isolates algorithmic randomness, variations in underlying library versions (such as NumPy) and local execution states can make significant differences in each team member's execution. If we were to scale or execute this project again, implementing containerized environments or strict dependency locks via a shared package manager would be non-negotiable. Ensuring an identical runtime architecture across all team members is fundamental to guaranteeing absolute scientific reproducibility.

6 Use of AI

Assistance of AI tools (Gemini/ChatGPT) was used for formatting LaTeX tables and code blocks and for text refinement.